



AI61002: Deep Learning Foundations and Applications

Module 5: Convolutional Neural Networks

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The Convolution Operation

- Convolutional neural networks (CNNs) comprise a class of neural networks that learn spatial hierarchies of features from spatial data, primarily for image/video processing and computer vision tasks.

- The basic building blocks of CNNs are convolutional filters, which perform a 2D convolution operation given by:

$$y(m, n) = x(m, n) * h(m, n) = \sum_i \sum_j x(i, j) h(m - i, n - j)$$

- For an $H \times W$ input and a $M \times N$ filter, the output of the convolution operation has the following sizes:
 - Valid output: $(H - M + 1) \times (W - N + 1)$
 - Full output: $(H + M - 1) \times (W + N - 1)$
- The convolution operation has the following properties:
 - Commutative: $x * h = h * x$
 - Associative: $h_1 * (h_2 * x) = (h_1 * h_2) * x$
 - Distributive: $(x_1 + x_2) * h = x_1 * h + x_2 * h$
- Convolution is shift invariant and spatially localized.

Convolutional Filters

- The building blocks of a CNN are the convolutional filters or kernels, which are used to perform convolutions between the input image or feature map, and the filter coefficients, learned from the data through back propagation.
- A convolutional filter is defined by the following hyperparameters:
 - *Kernel size*: spatial dimensions of the filter
 - *Stride*: number of steps by which the filter slides over the input
 - *Padding width*: number of extra rows and columns added along the boundary of the input

padding to compute response of the edges

0	0	0	0	0	0	0	0
0	4	9	2	5	8	3	0
0	5	6	2	4	0	3	0
0	2	4	5	4	5	2	0
0	5	6	5	4	7	8	0
0	5	7	7	9	2	1	0
0	5	8	5	3	8	4	0
0	0	0	0	0	0	0	0

1

0

-1

1

0

-1

1

0

-1

this kernel is edge detector

=

-15	6	

stride = 2 here

$$\lfloor W_o = \frac{W_i + 2P - K}{S} + 1 \rfloor$$

W_o : size of output feature map, W_i : size of input image/feature map
 P : padding size, K : filter/kernel size, S : stride

Convolutional Filters

- Padding the borders of the input image or feature map is used to control the size of the output feature map after convolution.
- Different types of padding could be applied, depending on the application.

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	5	3	8	1	6	0	0
0	0	4	7	2	9	0	0	0
0	0	3	5	7	8	1	0	0
0	0	2	6	0	4	3	0	0
0	0	9	1	5	2	7	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Zero/Constant Padding

edge effect
sharp discontinuity at the edge

5	5	5	3	8	1	6	6	6
5	5	5	3	8	1	6	6	6
5	5	5	3	8	1	6	6	6
4	4	4	7	2	9	0	0	0
3	3	3	5	7	8	1	1	1
2	2	2	6	0	4	3	3	3
9	9	9	1	5	2	7	7	7
9	9	9	1	5	2	7	7	7
9	9	9	1	5	2	7	7	7

Replicate/Edge Padding

this also has some edge effects, due to repetition of edge info
there is discontinuity wrt rest of the features of the inputs

7	4	4	7	2	9	0	0	9
3	5	5	3	8	1	6	6	1
3	5	5	3	8	1	6	6	1
7	4	4	7	2	9	0	0	9
5	3	3	5	7	8	1	1	8
6	2	2	6	0	4	3	3	4
1	9	9	1	5	2	7	7	2
1	9	9	1	5	2	7	7	2
6	2	2	6	0	4	3	3	4

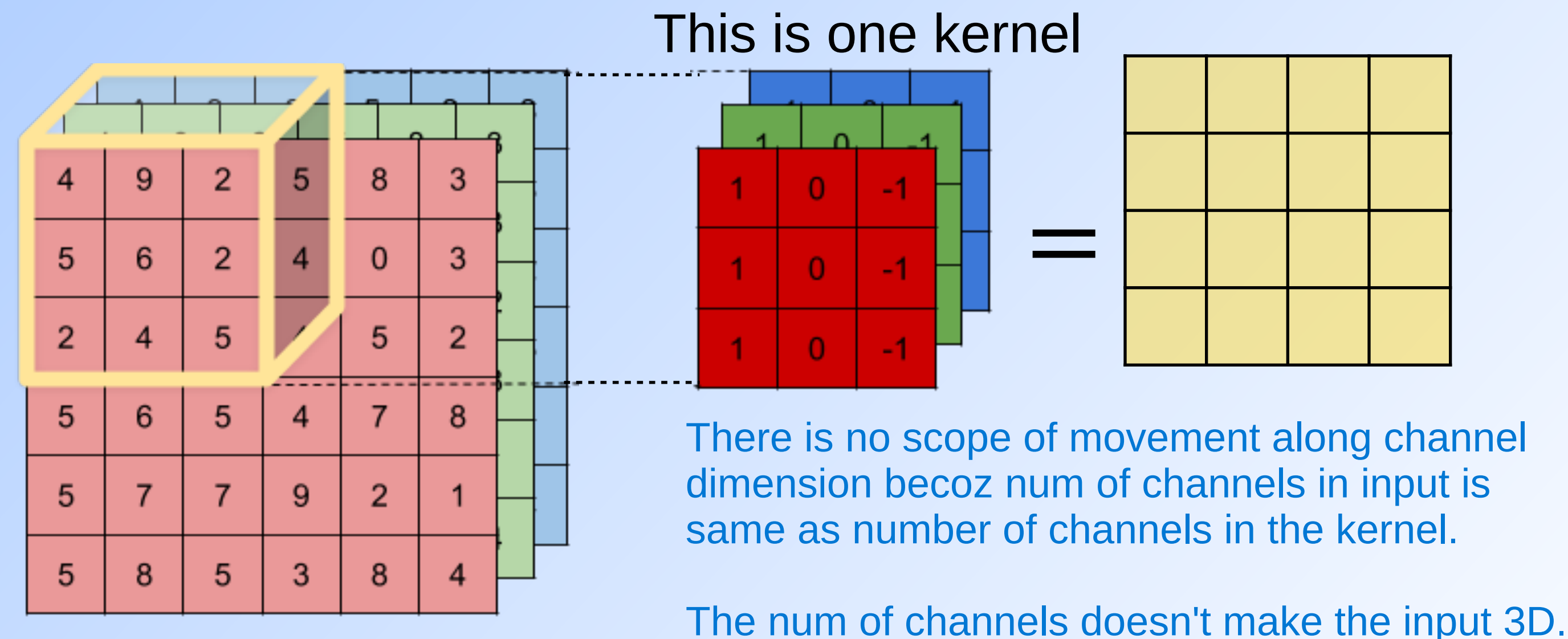
Reflective Padding

4	3	2	6	0	4	3	2	6
2	7	9	1	5	2	7	9	1
1	6	5	3	8	1	6	5	3
9	0	4	7	2	9	0	4	7
8	1	3	5	7	8	1	3	5
4	3	2	6	0	4	3	2	6
2	7	9	1	5	2	7	9	1
1	6	5	3	8	1	6	5	3
9	0	4	7	2	9	0	4	7

Circular Padding

Convolutional Filters

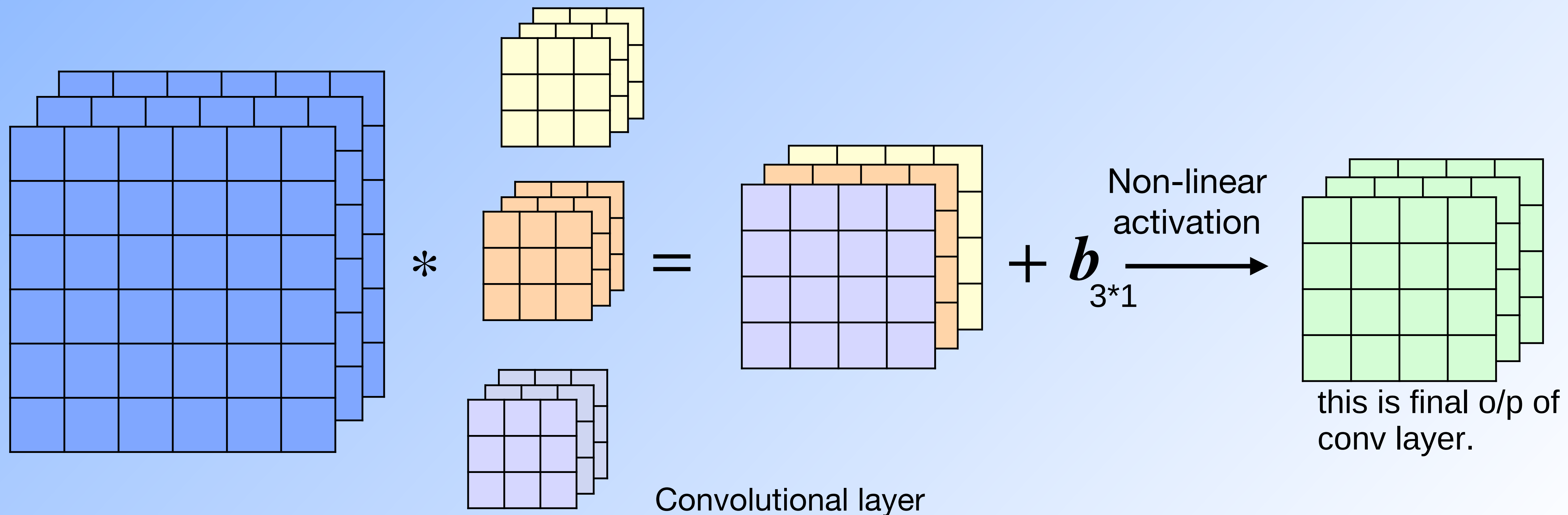
- The number of channels of the input must match the number of channels of the convolutional filter.
- The filters are applied channel-wise and the result of the convolution operation for each channel are summed to constitute the final output.
- The use of convolutional filters achieves the following:
 - Spatial invariance
 - Preservation of spatial structure
 - Parameter sharing
 - Sparsity of connections



spatial invariance :: regardless of the location of the obj it can detect it coz same set of weights are used for processing the image.
 Preservation of spatial structure :: Relative pos of features in the image are unaltered in the o/p, cat located left to dog in input will result in cat located to left of dog in the o/p, in reflective padding reflection of features occur at the edges
 parameter sharing entails same set of weights are used to process the entire input. 27 weights are used to process the entire input image, in FCNN each input will have its own weight and we would need 36×3
 sparsity of connections :: coz of less num of weights, o/p at any point will be determined by the local inputs not all the inputs due to sparse connections.

Convolutional Layer

- The output has multiple channels when multiple convolutional filters are used.
- A learnable bias might be added to the result obtained after the convolution operation, followed by the element-wise application of a non-linear activation function (such as ReLU) to constitute a convolutional layer



3 kernels here
Each kernel/neuron has a bias so bias dim = 3 X 1

Backpropagation Through Convolutional Filter

- The weights of the convolutional filters are updated using backpropagation using training
- Consider the following convolution operation between an input X and a filter W :

x_{11}	x_{12}	x_{13}	x_{14}
x_{21}	x_{22}	x_{23}	x_{24}
x_{31}	x_{32}	x_{33}	x_{34}
x_{41}	x_{42}	x_{43}	x_{44}

X

*

w_{11}	w_{12}	w_{13}
w_{21}	w_{22}	w_{23}
w_{31}	w_{32}	w_{33}

W

=

z_{11}	z_{12}
z_{21}	z_{22}

Z

Backpropagation Through Convolutional Filters

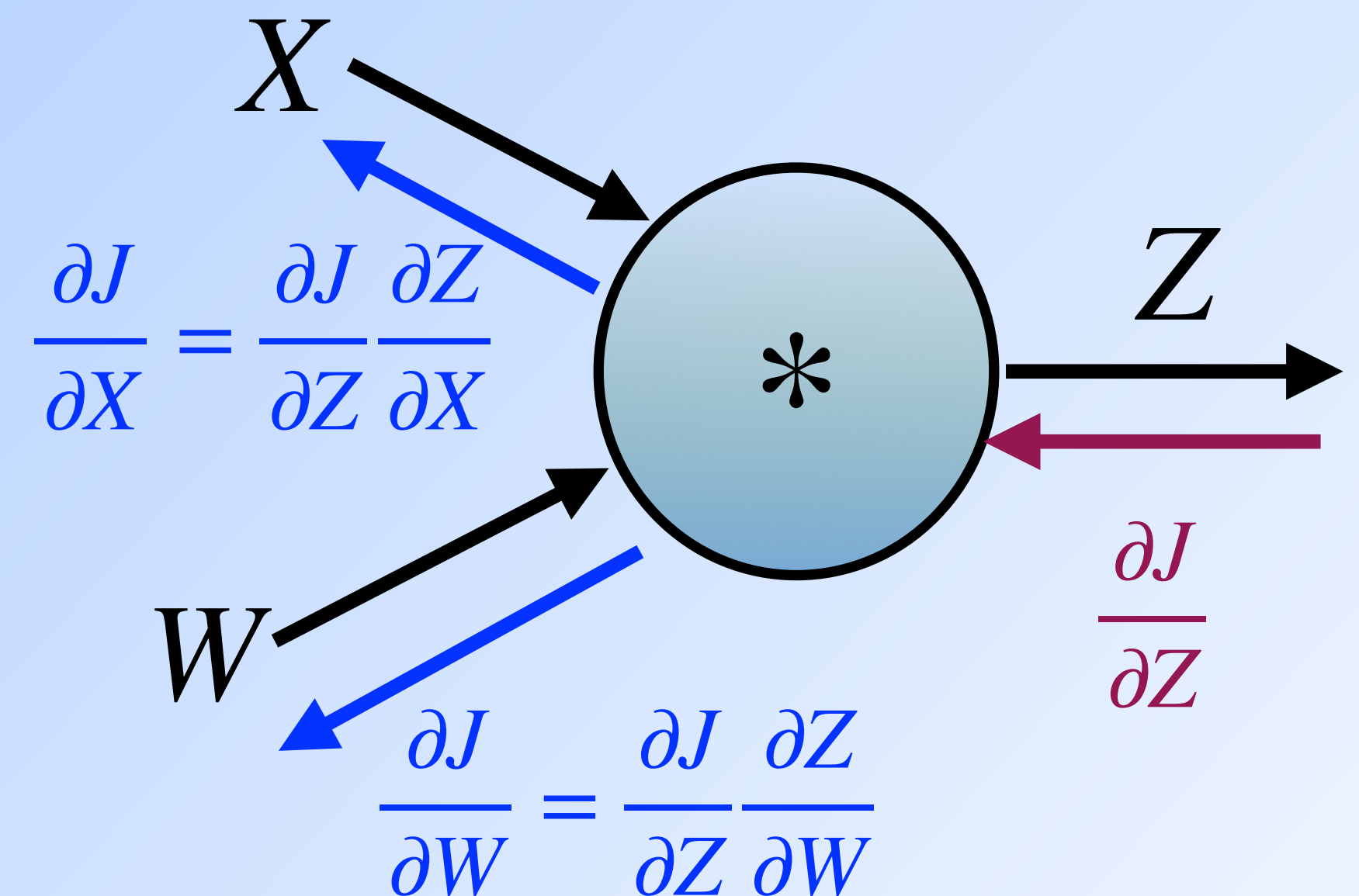
- The gradient of the cost function needs to be computed with respect to the input X and kernel W

Using chain rule, the gradient of the cost with respect to the convolutional filter weights could be computed as follows:

$$\frac{\partial J}{\partial W} = \frac{\partial J}{\partial Z} \frac{\partial Z}{\partial W}$$

For every element of W ,

$$\frac{\partial J}{\partial w_{ij}} = \sum_m \sum_n \frac{\partial J}{\partial z_{mn}} \frac{\partial z_{mn}}{\partial w_{ij}}$$



Backpropagation Through Convolutional Filters

$$z_{11} = x_{11}w_{11} + x_{12}w_{12} + x_{13}w_{13} + x_{21}w_{21} + x_{22}w_{22} + x_{23}w_{23} + x_{31}w_{31} + x_{32}w_{32} + x_{33}w_{33}$$

$$z_{12} = x_{12}w_{11} + x_{13}w_{12} + x_{14}w_{13} + x_{22}w_{21} + x_{23}w_{22} + x_{24}w_{23} + x_{32}w_{31} + x_{33}w_{32} + x_{34}w_{33}$$

$$z_{21} = x_{21}w_{11} + x_{22}w_{12} + x_{23}w_{13} + x_{31}w_{21} + x_{32}w_{22} + x_{33}w_{23} + x_{41}w_{31} + x_{42}w_{32} + x_{43}w_{33}$$

$$z_{22} = x_{22}w_{11} + x_{23}w_{12} + x_{24}w_{13} + x_{32}w_{21} + x_{33}w_{22} + x_{34}w_{23} + x_{42}w_{31} + x_{43}w_{32} + x_{44}w_{33}$$

$$\frac{\partial J}{\partial w_{11}} = \frac{\partial J}{\partial z_{11}} \frac{\partial z_{11}}{\partial w_{11}} + \frac{\partial J}{\partial z_{12}} \frac{\partial z_{12}}{\partial w_{11}} + \frac{\partial J}{\partial z_{21}} \frac{\partial z_{21}}{\partial w_{11}} + \frac{\partial J}{\partial z_{22}} \frac{\partial z_{22}}{\partial w_{11}} = \frac{\partial J}{\partial z_{11}} x_{11} + \frac{\partial J}{\partial z_{12}} x_{12} + \frac{\partial J}{\partial z_{21}} x_{21} + \frac{\partial J}{\partial z_{22}} x_{22}$$

$$\frac{\partial J}{\partial w_{12}} = \frac{\partial J}{\partial z_{11}} \frac{\partial z_{11}}{\partial w_{12}} + \frac{\partial J}{\partial z_{12}} \frac{\partial z_{12}}{\partial w_{12}} + \frac{\partial J}{\partial z_{21}} \frac{\partial z_{21}}{\partial w_{12}} + \frac{\partial J}{\partial z_{22}} \frac{\partial z_{22}}{\partial w_{12}} = \frac{\partial J}{\partial z_{11}} x_{12} + \frac{\partial J}{\partial z_{12}} x_{13} + \frac{\partial J}{\partial z_{21}} x_{22} + \frac{\partial J}{\partial z_{22}} x_{23}$$

$$\frac{\partial J}{\partial w_{13}} = \frac{\partial J}{\partial z_{11}} \frac{\partial z_{11}}{\partial w_{13}} + \frac{\partial J}{\partial z_{12}} \frac{\partial z_{12}}{\partial w_{13}} + \frac{\partial J}{\partial z_{21}} \frac{\partial z_{21}}{\partial w_{13}} + \frac{\partial J}{\partial z_{22}} \frac{\partial z_{22}}{\partial w_{13}} = \frac{\partial J}{\partial z_{11}} x_{13} + \frac{\partial J}{\partial z_{12}} x_{14} + \frac{\partial J}{\partial z_{21}} x_{23} + \frac{\partial J}{\partial z_{22}} x_{24}$$

Backpropagation Through Convolutional Filter

$$z_{11} = x_{11}w_{11} + x_{12}w_{12} + x_{13}w_{13} + x_{21}w_{21} + x_{22}w_{22} + x_{23}w_{23} + x_{31}w_{31} + x_{32}w_{32} + x_{33}w_{33}$$

$$z_{12} = x_{12}w_{11} + x_{13}w_{12} + x_{14}w_{13} + x_{22}w_{21} + x_{23}w_{22} + x_{24}w_{23} + x_{32}w_{31} + x_{33}w_{32} + x_{34}w_{33}$$

$$z_{21} = x_{21}w_{11} + x_{22}w_{12} + x_{23}w_{13} + x_{31}w_{21} + x_{32}w_{22} + x_{33}w_{23} + x_{41}w_{31} + x_{42}w_{32} + x_{43}w_{33}$$

$$z_{22} = x_{22}w_{11} + x_{23}w_{12} + x_{24}w_{13} + x_{32}w_{21} + x_{33}w_{22} + x_{34}w_{23} + x_{42}w_{31} + x_{43}w_{32} + x_{44}w_{33}$$

$$\frac{\partial J}{\partial w_{21}} = \frac{\partial J}{\partial z_{11}} \frac{\partial z_{11}}{\partial w_{21}} + \frac{\partial J}{\partial z_{12}} \frac{\partial z_{12}}{\partial w_{21}} + \frac{\partial J}{\partial z_{21}} \frac{\partial z_{21}}{\partial w_{21}} + \frac{\partial J}{\partial z_{22}} \frac{\partial z_{22}}{\partial w_{21}} = \frac{\partial J}{\partial z_{11}} x_{21} + \frac{\partial J}{\partial z_{12}} x_{22} + \frac{\partial J}{\partial z_{21}} x_{31} + \frac{\partial J}{\partial z_{22}} x_{32}$$

$$\frac{\partial J}{\partial w_{22}} = \frac{\partial J}{\partial z_{11}} \frac{\partial z_{11}}{\partial w_{22}} + \frac{\partial J}{\partial z_{12}} \frac{\partial z_{12}}{\partial w_{22}} + \frac{\partial J}{\partial z_{21}} \frac{\partial z_{21}}{\partial w_{22}} + \frac{\partial J}{\partial z_{22}} \frac{\partial z_{22}}{\partial w_{22}} = \frac{\partial J}{\partial z_{11}} x_{22} + \frac{\partial J}{\partial z_{12}} x_{23} + \frac{\partial J}{\partial z_{21}} x_{32} + \frac{\partial J}{\partial z_{22}} x_{33}$$

$$\frac{\partial J}{\partial w_{23}} = \frac{\partial J}{\partial z_{11}} \frac{\partial z_{11}}{\partial w_{23}} + \frac{\partial J}{\partial z_{12}} \frac{\partial z_{12}}{\partial w_{23}} + \frac{\partial J}{\partial z_{21}} \frac{\partial z_{21}}{\partial w_{23}} + \frac{\partial J}{\partial z_{22}} \frac{\partial z_{22}}{\partial w_{23}} = \frac{\partial J}{\partial z_{11}} x_{23} + \frac{\partial J}{\partial z_{12}} x_{24} + \frac{\partial J}{\partial z_{21}} x_{33} + \frac{\partial J}{\partial z_{22}} x_{34}$$

Backpropagation Through Convolutional Filters

$$z_{11} = x_{11}w_{11} + x_{12}w_{12} + x_{13}w_{13} + x_{21}w_{21} + x_{22}w_{22} + x_{23}w_{23} + x_{31}w_{31} + x_{32}w_{32} + x_{33}w_{33}$$

$$z_{12} = x_{12}w_{11} + x_{13}w_{12} + x_{14}w_{13} + x_{22}w_{21} + x_{23}w_{22} + x_{24}w_{23} + x_{32}w_{31} + x_{33}w_{32} + x_{34}w_{33}$$

$$z_{21} = x_{21}w_{11} + x_{22}w_{12} + x_{23}w_{13} + x_{31}w_{21} + x_{32}w_{22} + x_{33}w_{23} + x_{41}w_{31} + x_{42}w_{32} + x_{43}w_{33}$$

$$z_{22} = x_{22}w_{11} + x_{23}w_{12} + x_{24}w_{13} + x_{32}w_{21} + x_{33}w_{22} + x_{34}w_{23} + x_{42}w_{31} + x_{43}w_{32} + x_{44}w_{33}$$

$$\frac{\partial J}{\partial w_{31}} = \frac{\partial J}{\partial z_{11}} \frac{\partial z_{11}}{\partial w_{31}} + \frac{\partial J}{\partial z_{12}} \frac{\partial z_{12}}{\partial w_{31}} + \frac{\partial J}{\partial z_{21}} \frac{\partial z_{21}}{\partial w_{31}} + \frac{\partial J}{\partial z_{22}} \frac{\partial z_{22}}{\partial w_{31}} = \frac{\partial J}{\partial z_{11}} x_{31} + \frac{\partial J}{\partial z_{12}} x_{32} + \frac{\partial J}{\partial z_{21}} x_{41} + \frac{\partial J}{\partial z_{22}} x_{42}$$

$$\frac{\partial J}{\partial w_{32}} = \frac{\partial J}{\partial z_{11}} \frac{\partial z_{11}}{\partial w_{32}} + \frac{\partial J}{\partial z_{12}} \frac{\partial z_{12}}{\partial w_{32}} + \frac{\partial J}{\partial z_{21}} \frac{\partial z_{21}}{\partial w_{32}} + \frac{\partial J}{\partial z_{22}} \frac{\partial z_{22}}{\partial w_{32}} = \frac{\partial J}{\partial z_{11}} x_{32} + \frac{\partial J}{\partial z_{12}} x_{33} + \frac{\partial J}{\partial z_{21}} x_{42} + \frac{\partial J}{\partial z_{22}} x_{43}$$

$$\frac{\partial J}{\partial w_{33}} = \frac{\partial J}{\partial z_{11}} \frac{\partial z_{11}}{\partial w_{33}} + \frac{\partial J}{\partial z_{12}} \frac{\partial z_{12}}{\partial w_{33}} + \frac{\partial J}{\partial z_{21}} \frac{\partial z_{21}}{\partial w_{33}} + \frac{\partial J}{\partial z_{22}} \frac{\partial z_{22}}{\partial w_{33}} = \frac{\partial J}{\partial z_{11}} x_{33} + \frac{\partial J}{\partial z_{12}} x_{34} + \frac{\partial J}{\partial z_{21}} x_{43} + \frac{\partial J}{\partial z_{22}} x_{44}$$

Backpropagation Through Convolutional Filters

The equations derived for the gradients of the cost w.r.t the weights of the convolutional kernel imply that:

x_{11}	x_{12}	x_{13}	x_{14}
x_{21}	x_{22}	x_{23}	x_{24}
x_{31}	x_{32}	x_{33}	x_{34}
x_{41}	x_{42}	x_{43}	x_{44}

 $*$

$\frac{\partial J}{\partial z_{11}}$	$\frac{\partial J}{\partial z_{12}}$
$\frac{\partial J}{\partial z_{21}}$	$\frac{\partial J}{\partial z_{22}}$

 $=$

$\frac{\partial J}{\partial w_{11}}$	$\frac{\partial J}{\partial w_{12}}$	$\frac{\partial J}{\partial w_{13}}$
$\frac{\partial J}{\partial w_{21}}$	$\frac{\partial J}{\partial w_{22}}$	$\frac{\partial J}{\partial w_{23}}$
$\frac{\partial J}{\partial w_{31}}$	$\frac{\partial J}{\partial w_{32}}$	$\frac{\partial J}{\partial w_{33}}$

$$\therefore \frac{\partial J}{\partial W} = X * \frac{\partial J}{\partial Z}$$

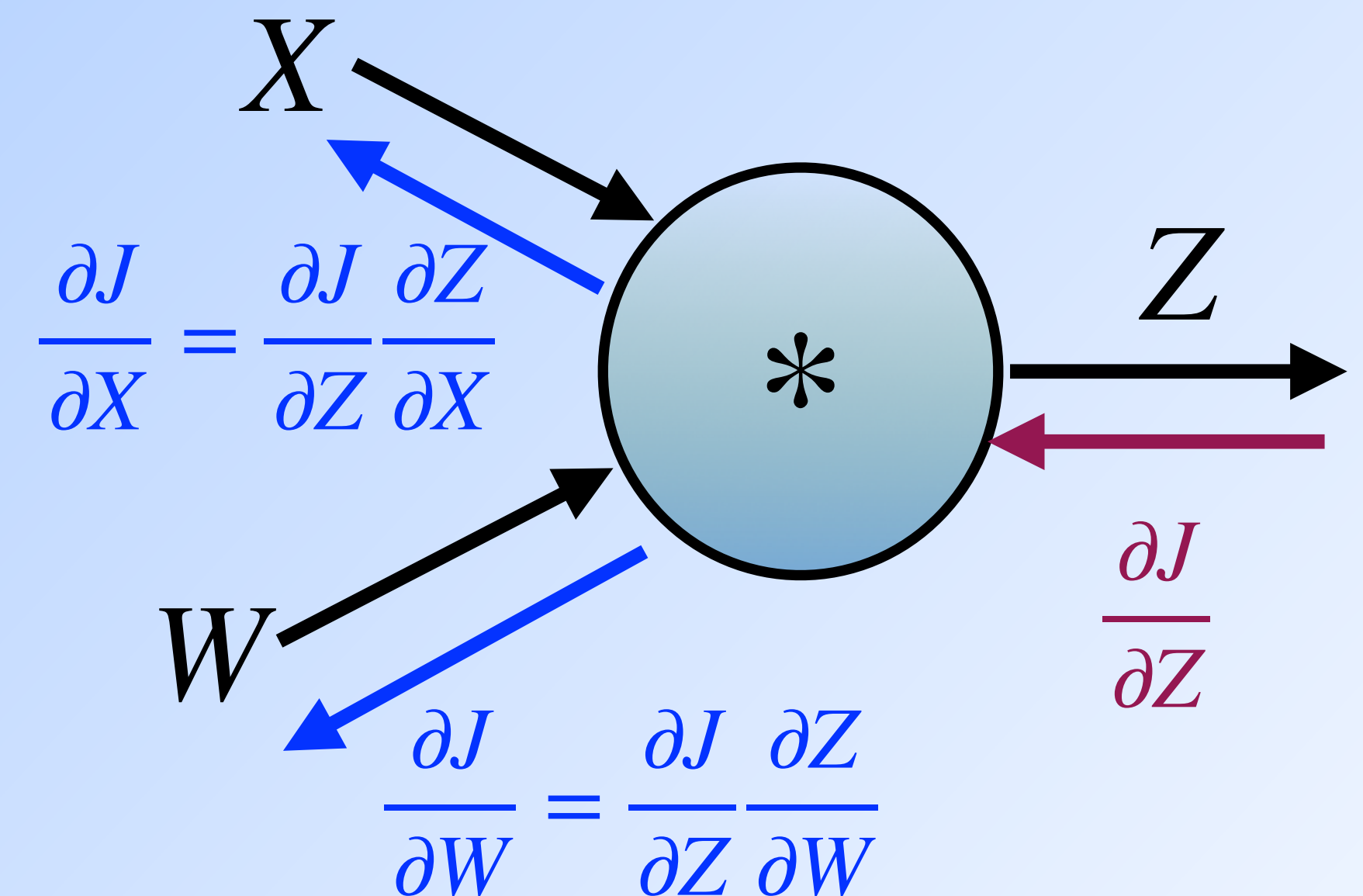
Backpropagation Through Convolutional Filters

Similarly, the gradient of the cost function with respect to the input X of the convolutional filter can be derived as follows:

$$\frac{\partial J}{\partial X} = \frac{\partial J}{\partial Z} \frac{\partial Z}{\partial X}$$

For each element of X ,

$$\frac{\partial J}{\partial x_{ij}} = \sum_m \sum_n \frac{\partial J}{\partial z_{mn}} \frac{\partial z_{mn}}{\partial x_{ij}}$$



Backpropagation Through Convolutional Filters

$$\frac{\partial J}{\partial x_{11}} = \frac{\partial J}{\partial z_{11}} w_{11}, \frac{\partial J}{\partial x_{12}} = \frac{\partial J}{\partial z_{11}} w_{12} + \frac{\partial J}{\partial z_{12}} w_{11}, \frac{\partial J}{\partial x_{13}} = \frac{\partial J}{\partial z_{11}} w_{13} + \frac{\partial J}{\partial z_{12}} w_{12}, \frac{\partial J}{\partial x_{14}} = \frac{\partial J}{\partial z_{12}} w_{13}$$

$$\frac{\partial J}{\partial x_{21}} = \frac{\partial J}{\partial z_{11}} w_{21} + \frac{\partial J}{\partial z_{21}} w_{11}, \frac{\partial J}{\partial x_{22}} = \frac{\partial J}{\partial z_{11}} w_{22} + \frac{\partial J}{\partial z_{12}} w_{21} + \frac{\partial J}{\partial z_{21}} w_{12} + \frac{\partial J}{\partial z_{22}} w_{11}, \frac{\partial J}{\partial x_{23}} = \frac{\partial J}{\partial z_{11}} w_{23} + \frac{\partial J}{\partial z_{12}} w_{22} + \frac{\partial J}{\partial z_{21}} w_{13} + \frac{\partial J}{\partial z_{22}} w_{12},$$

$$\frac{\partial J}{\partial x_{24}} = \frac{\partial J}{\partial z_{12}} w_{23} + \frac{\partial J}{\partial z_{22}} w_{13}$$

$$\frac{\partial J}{\partial x_{31}} = \frac{\partial J}{\partial z_{11}} w_{31} + \frac{\partial J}{\partial z_{21}} w_{21}, \frac{\partial J}{\partial x_{32}} = \frac{\partial J}{\partial z_{11}} w_{32} + \frac{\partial J}{\partial z_{12}} w_{31} + \frac{\partial J}{\partial z_{21}} w_{22} + \frac{\partial J}{\partial z_{22}} w_{21}, \frac{\partial J}{\partial x_{33}} = \frac{\partial J}{\partial z_{11}} w_{33} + \frac{\partial J}{\partial z_{12}} w_{32} + \frac{\partial J}{\partial z_{21}} w_{23} + \frac{\partial J}{\partial z_{22}} w_{22},$$

$$\frac{\partial J}{\partial x_{34}} = \frac{\partial J}{\partial z_{12}} w_{33} + \frac{\partial J}{\partial z_{22}} w_{23}$$

$$\frac{\partial J}{\partial x_{41}} = \frac{\partial J}{\partial z_{21}} w_{31}, \frac{\partial J}{\partial x_{42}} = \frac{\partial J}{\partial z_{21}} w_{32} + \frac{\partial J}{\partial z_{22}} w_{31}, \frac{\partial J}{\partial x_{43}} = \frac{\partial J}{\partial z_{21}} w_{33} + \frac{\partial J}{\partial z_{22}} w_{32}, \frac{\partial J}{\partial x_{44}} = \frac{\partial J}{\partial z_{22}} w_{33}.$$

Backpropagation Through Convolutional Filters

- The gradients of the cost with respect to the input X can be expressed as the full convolution result of the filter W with the 180° rotated version of the upstream gradient matrix $\frac{\partial J}{\partial Z}$, i.e.:

w_{11}	w_{12}	w_{13}
w_{21}	w_{22}	w_{23}
w_{31}	w_{32}	w_{33}

 $*$

$\frac{\partial J}{\partial z_{22}}$	$\frac{\partial J}{\partial z_{21}}$
$\frac{\partial J}{\partial z_{12}}$	$\frac{\partial J}{\partial z_{11}}$

 $=$

$\frac{\partial J}{\partial x_{11}}$	$\frac{\partial J}{\partial x_{12}}$	$\frac{\partial J}{\partial x_{13}}$	$\frac{\partial J}{\partial x_{14}}$
$\frac{\partial J}{\partial x_{21}}$	$\frac{\partial J}{\partial x_{22}}$	$\frac{\partial J}{\partial x_{23}}$	$\frac{\partial J}{\partial x_{24}}$
$\frac{\partial J}{\partial x_{31}}$	$\frac{\partial J}{\partial x_{32}}$	$\frac{\partial J}{\partial x_{33}}$	$\frac{\partial J}{\partial x_{34}}$
$\frac{\partial J}{\partial x_{41}}$	$\frac{\partial J}{\partial x_{42}}$	$\frac{\partial J}{\partial x_{43}}$	$\frac{\partial J}{\partial x_{44}}$

$$\therefore \frac{\partial J}{\partial X} = W * \overline{\frac{\partial J}{\partial Z}}, \text{ where } \overline{\frac{\partial J}{\partial Z}} \text{ is } \frac{\partial J}{\partial Z} \text{ rotated by } 180^\circ.$$

Backpropagation Through Convolutional Filters

$\frac{\partial J}{\partial z_{22}}$	$\frac{\partial J}{\partial z_{21}}$		
$\frac{\partial J}{\partial z_{12}}$	$\frac{\partial J}{\partial z_{11}} w_{11}$	w_{12}	w_{13}
	w_{21}	w_{22}	w_{23}
	w_{31}	w_{32}	w_{33}

$$\frac{\partial J}{\partial x_{11}} = \frac{\partial J}{\partial z_{11}} w_{11}$$

$\frac{\partial J}{\partial z_{22}}$	$\frac{\partial J}{\partial z_{21}}$	
$\frac{\partial J}{\partial z_{12}} w_{11}$	$\frac{\partial J}{\partial z_{11}} w_{12}$	w_{13}
w_{21}	w_{22}	w_{23}
w_{31}	w_{32}	w_{33}

$$\frac{\partial J}{\partial x_{12}} = \frac{\partial J}{\partial z_{11}} w_{12} + \frac{\partial J}{\partial z_{12}} w_{11}$$

	$\frac{\partial J}{\partial z_{22}}$	$\frac{\partial J}{\partial z_{21}}$
w_{11}	$\frac{\partial J}{\partial z_{12}} w_{12}$	$\frac{\partial J}{\partial z_{11}} w_{13}$
w_{21}	w_{22}	w_{23}
w_{31}	w_{32}	w_{33}

$$\frac{\partial J}{\partial x_{13}} = \frac{\partial J}{\partial z_{11}} w_{13} + \frac{\partial J}{\partial z_{12}} w_{12}$$

		$\frac{\partial J}{\partial z_{22}}$	$\frac{\partial J}{\partial z_{21}}$
w_{11}	w_{12}	$w_{13} \frac{\partial J}{\partial z_{12}}$	$\frac{\partial J}{\partial z_{11}}$
w_{21}	w_{22}	w_{23}	
w_{31}	w_{32}	w_{33}	

$$\frac{\partial J}{\partial x_{14}} = \frac{\partial J}{\partial z_{12}} w_{13}$$

$\frac{\partial J}{\partial z_{22}}$	$\frac{\partial J}{\partial z_{21}} w_{11}$	w_{12}	w_{13}
$\frac{\partial J}{\partial z_{12}}$	$\frac{\partial J}{\partial z_{11}} w_{21}$	w_{22}	w_{23}
	w_{31}	w_{32}	w_{33}

$$\frac{\partial J}{\partial x_{21}} = \frac{\partial J}{\partial z_{11}} w_{21} + \frac{\partial J}{\partial z_{21}} w_{11}$$

$\frac{\partial J}{\partial z_{22}} w_{11}$	$\frac{\partial J}{\partial z_{21}} w_{12}$	w_{13}
$\frac{\partial J}{\partial z_{12}} w_{21}$	$\frac{\partial J}{\partial z_{11}} w_{22}$	w_{23}
w_{31}	w_{32}	w_{33}

$$\frac{\partial J}{\partial x_{22}} = \frac{\partial J}{\partial z_{11}} w_{22} + \frac{\partial J}{\partial z_{12}} w_{21} + \frac{\partial J}{\partial z_{21}} w_{12} + \frac{\partial J}{\partial z_{22}} w_{11}$$

w_{11}	$\frac{\partial J}{\partial z_{22}} w_{12}$	$\frac{\partial J}{\partial z_{21}} w_{13}$
w_{21}	$\frac{\partial J}{\partial z_{12}} w_{22}$	$\frac{\partial J}{\partial z_{11}} w_{23}$
w_{31}	w_{32}	w_{33}

$$\frac{\partial J}{\partial x_{23}} = \frac{\partial J}{\partial z_{11}} w_{23} + \frac{\partial J}{\partial z_{12}} w_{22} + \frac{\partial J}{\partial z_{21}} w_{13} + \frac{\partial J}{\partial z_{22}} w_{12}$$

w_{11}	w_{12}	$\frac{\partial J}{\partial z_{22}} w_{13}$	$\frac{\partial J}{\partial z_{21}}$
w_{21}	w_{22}	$\frac{\partial J}{\partial z_{12}} w_{23}$	$\frac{\partial J}{\partial z_{11}}$
w_{31}	w_{32}	w_{33}	

$$\frac{\partial J}{\partial x_{24}} = \frac{\partial J}{\partial z_{12}} w_{23} + \frac{\partial J}{\partial z_{22}} w_{13}$$

Backpropagation Through Convolutional Filter

- Now consider a strided convolution with stride=2:

x_{11}	x_{12}	x_{13}	x_{14}
x_{21}	x_{22}	x_{23}	x_{24}
x_{31}	x_{32}	x_{33}	x_{34}
x_{41}	x_{42}	x_{43}	x_{44}

X

*

w_{11}	w_{12}
w_{21}	w_{22}

W

=

z_{11}	z_{12}
z_{21}	z_{22}

Z

$$z_{11} = x_{11}w_{11} + x_{12}w_{12} + x_{21}w_{21} + x_{22}w_{22}$$

$$z_{12} = x_{13}w_{11} + x_{14}w_{12} + x_{23}w_{21} + x_{24}w_{22}$$

$$z_{21} = x_{31}w_{11} + x_{32}w_{12} + x_{41}w_{21} + x_{42}w_{22}$$

$$z_{22} = x_{33}w_{11} + x_{34}w_{12} + x_{43}w_{21} + x_{44}w_{22}$$

Backpropagation Through Convolutional Filter

$$\begin{aligned}
 \frac{\partial J}{\partial w_{11}} &= \frac{\partial J}{\partial z_{11}} \frac{\partial z_{11}}{\partial w_{11}} + \frac{\partial J}{\partial z_{12}} \frac{\partial z_{12}}{\partial w_{11}} + \frac{\partial J}{\partial z_{21}} \frac{\partial z_{21}}{\partial w_{11}} + \frac{\partial J}{\partial z_{22}} \frac{\partial z_{22}}{\partial w_{11}} = \frac{\partial J}{\partial z_{11}} x_{11} + \frac{\partial J}{\partial z_{12}} x_{13} + \frac{\partial J}{\partial z_{21}} x_{31} + \frac{\partial J}{\partial z_{22}} x_{33} \\
 \frac{\partial J}{\partial w_{12}} &= \frac{\partial J}{\partial z_{11}} \frac{\partial z_{11}}{\partial w_{12}} + \frac{\partial J}{\partial z_{12}} \frac{\partial z_{12}}{\partial w_{12}} + \frac{\partial J}{\partial z_{21}} \frac{\partial z_{21}}{\partial w_{12}} + \frac{\partial J}{\partial z_{22}} \frac{\partial z_{22}}{\partial w_{12}} = \frac{\partial J}{\partial z_{11}} x_{12} + \frac{\partial J}{\partial z_{12}} x_{14} + \frac{\partial J}{\partial z_{21}} x_{32} + \frac{\partial J}{\partial z_{22}} x_{34} \\
 \frac{\partial J}{\partial w_{21}} &= \frac{\partial J}{\partial z_{11}} \frac{\partial z_{11}}{\partial w_{21}} + \frac{\partial J}{\partial z_{12}} \frac{\partial z_{12}}{\partial w_{21}} + \frac{\partial J}{\partial z_{21}} \frac{\partial z_{21}}{\partial w_{21}} + \frac{\partial J}{\partial z_{22}} \frac{\partial z_{22}}{\partial w_{21}} = \frac{\partial J}{\partial z_{11}} x_{21} + \frac{\partial J}{\partial z_{12}} x_{23} + \frac{\partial J}{\partial z_{21}} x_{41} + \frac{\partial J}{\partial z_{22}} x_{43} \\
 \frac{\partial J}{\partial w_{22}} &= \frac{\partial J}{\partial z_{11}} \frac{\partial z_{11}}{\partial w_{22}} + \frac{\partial J}{\partial z_{12}} \frac{\partial z_{12}}{\partial w_{22}} + \frac{\partial J}{\partial z_{21}} \frac{\partial z_{21}}{\partial w_{22}} + \frac{\partial J}{\partial z_{22}} \frac{\partial z_{22}}{\partial w_{22}} = \frac{\partial J}{\partial z_{11}} x_{22} + \frac{\partial J}{\partial z_{12}} x_{24} + \frac{\partial J}{\partial z_{21}} x_{42} + \frac{\partial J}{\partial z_{22}} x_{44}
 \end{aligned}$$

Backpropagation Through Convolutional Filter

- The gradients of the cost function with respect to the weights can thus be computed using a dilated convolution between the input X and the upstream gradient $\frac{\partial J}{\partial Z}$, dilated by a factor equal to the stride.

x_{11}	x_{12}	x_{13}	x_{14}
x_{21}	x_{22}	x_{23}	x_{24}
x_{31}	x_{32}	x_{33}	x_{34}
x_{41}	x_{42}	x_{43}	x_{44}

*

$\frac{\partial J}{\partial z_{11}}$		$\frac{\partial J}{\partial z_{12}}$
$\frac{\partial J}{\partial z_{21}}$		$\frac{\partial J}{\partial z_{22}}$

=

$\frac{\partial J}{\partial w_{11}}$	$\frac{\partial J}{\partial w_{12}}$
$\frac{\partial J}{\partial w_{21}}$	$\frac{\partial J}{\partial w_{22}}$

$$\therefore \frac{\partial J}{\partial W} = X * \widetilde{\frac{\partial J}{\partial Z}}, \text{ where } \widetilde{\frac{\partial J}{\partial Z}} \text{ is the dilated version of } \frac{\partial J}{\partial Z}.$$

Backpropagation Through Convolutional Filter

$$\begin{aligned}
 \frac{\partial J}{\partial x_{11}} &= \frac{\partial J}{\partial z_{11}} w_{11}, & \frac{\partial J}{\partial x_{12}} &= \frac{\partial J}{\partial z_{11}} w_{12}, & \frac{\partial J}{\partial x_{13}} &= \frac{\partial J}{\partial z_{12}} w_{11}, & \frac{\partial J}{\partial x_{14}} &= \frac{\partial J}{\partial z_{12}} w_{12} \\
 \frac{\partial J}{\partial x_{21}} &= \frac{\partial J}{\partial z_{11}} w_{21}, & \frac{\partial J}{\partial x_{22}} &= \frac{\partial J}{\partial z_{11}} w_{22}, & \frac{\partial J}{\partial x_{23}} &= \frac{\partial J}{\partial z_{12}} w_{21}, & \frac{\partial J}{\partial x_{24}} &= \frac{\partial J}{\partial z_{12}} w_{22} \\
 \frac{\partial J}{\partial x_{31}} &= \frac{\partial J}{\partial z_{21}} w_{11}, & \frac{\partial J}{\partial x_{32}} &= \frac{\partial J}{\partial z_{21}} w_{22}, & \frac{\partial J}{\partial x_{33}} &= \frac{\partial J}{\partial z_{22}} w_{11}, & \frac{\partial J}{\partial x_{34}} &= \frac{\partial J}{\partial z_{22}} w_{12} \\
 \frac{\partial J}{\partial x_{41}} &= \frac{\partial J}{\partial z_{21}} w_{21}, & \frac{\partial J}{\partial x_{42}} &= \frac{\partial J}{\partial z_{21}} w_{22}, & \frac{\partial J}{\partial x_{43}} &= \frac{\partial J}{\partial z_{22}} w_{21}, & \frac{\partial J}{\partial x_{44}} &= \frac{\partial J}{\partial z_{22}} w_{22}
 \end{aligned}$$

Backpropagation Through Convolutional Filters

- The gradients of the cost with respect to the input X can be expressed as the full convolution result of the filter W with the 180° rotated and dilated (by a factor of the stride) version of the upstream gradient matrix $\frac{\partial J}{\partial Z}$, i.e.:

w_{11}	w_{12}
w_{21}	w_{22}

*

$\frac{\partial J}{\partial z_{22}}$		$\frac{\partial J}{\partial z_{21}}$
$\frac{\partial J}{\partial z_{12}}$		$\frac{\partial J}{\partial z_{11}}$

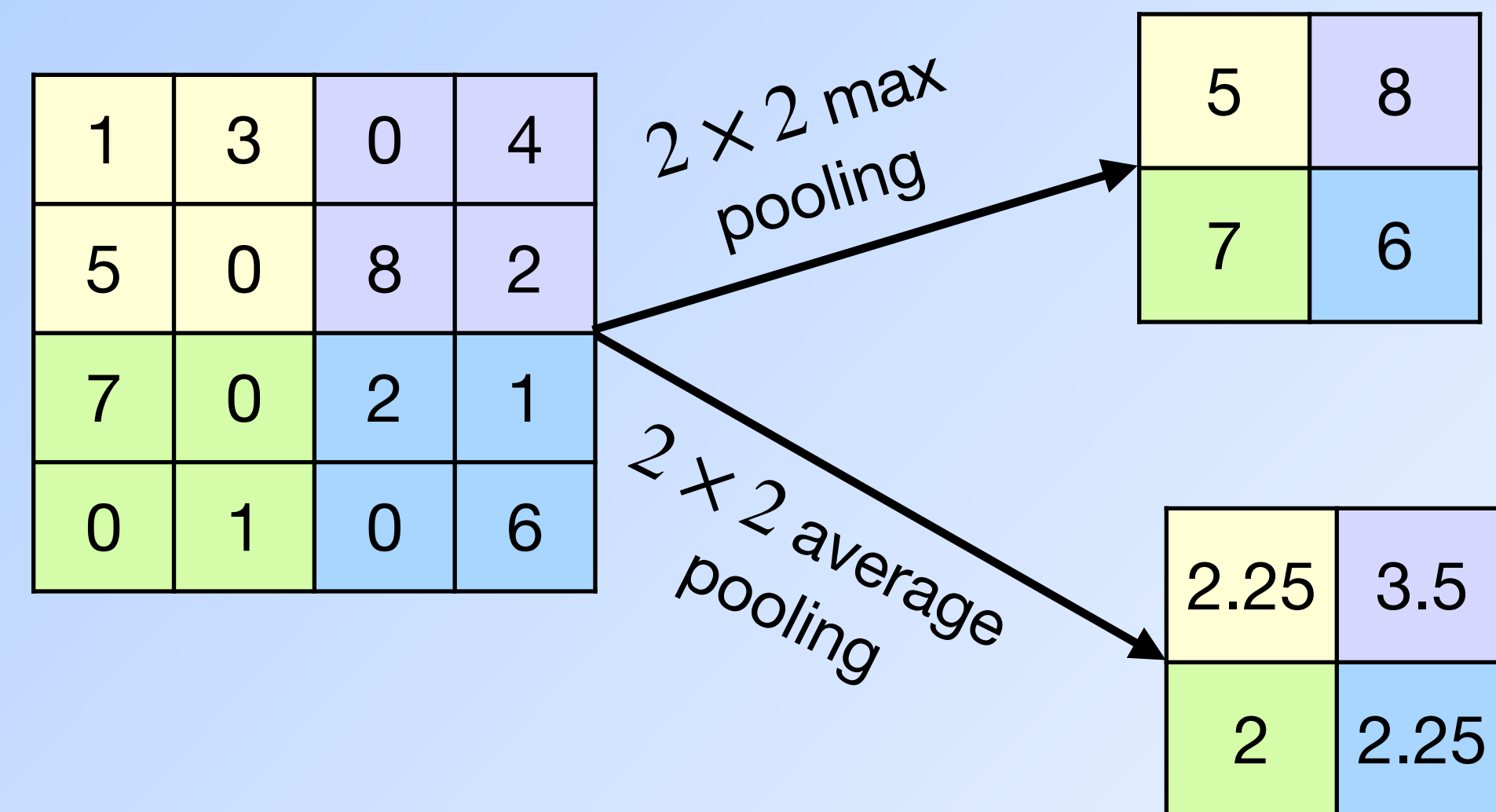
=

$\frac{\partial J}{\partial x_{11}}$	$\frac{\partial J}{\partial x_{12}}$	$\frac{\partial J}{\partial x_{13}}$	$\frac{\partial J}{\partial x_{14}}$
$\frac{\partial J}{\partial x_{21}}$	$\frac{\partial J}{\partial x_{22}}$	$\frac{\partial J}{\partial x_{23}}$	$\frac{\partial J}{\partial x_{24}}$
$\frac{\partial J}{\partial x_{31}}$	$\frac{\partial J}{\partial x_{32}}$	$\frac{\partial J}{\partial x_{33}}$	$\frac{\partial J}{\partial x_{34}}$
$\frac{\partial J}{\partial x_{41}}$	$\frac{\partial J}{\partial x_{42}}$	$\frac{\partial J}{\partial x_{43}}$	$\frac{\partial J}{\partial x_{44}}$

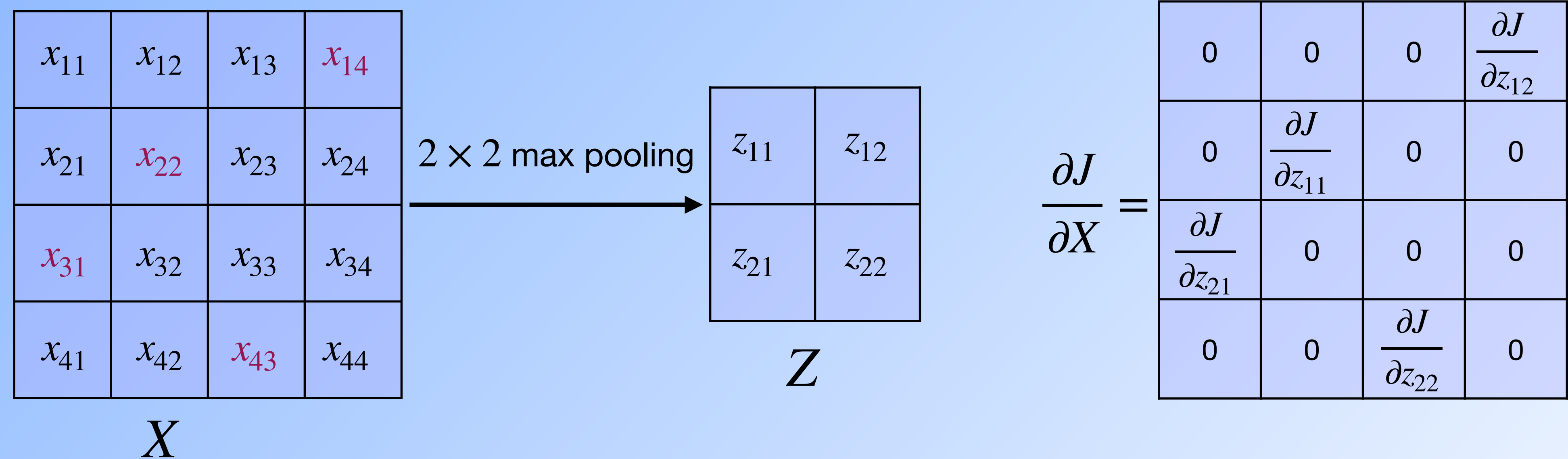
$$\therefore \frac{\partial J}{\partial X} = W * \overline{\frac{\partial J}{\partial Z}}, \text{ where } \overline{\frac{\partial J}{\partial Z}} \text{ is } \frac{\partial J}{\partial Z} \text{ rotated by } 180^\circ \text{ and dilated by a factor of 2.}$$

Pooling Layer

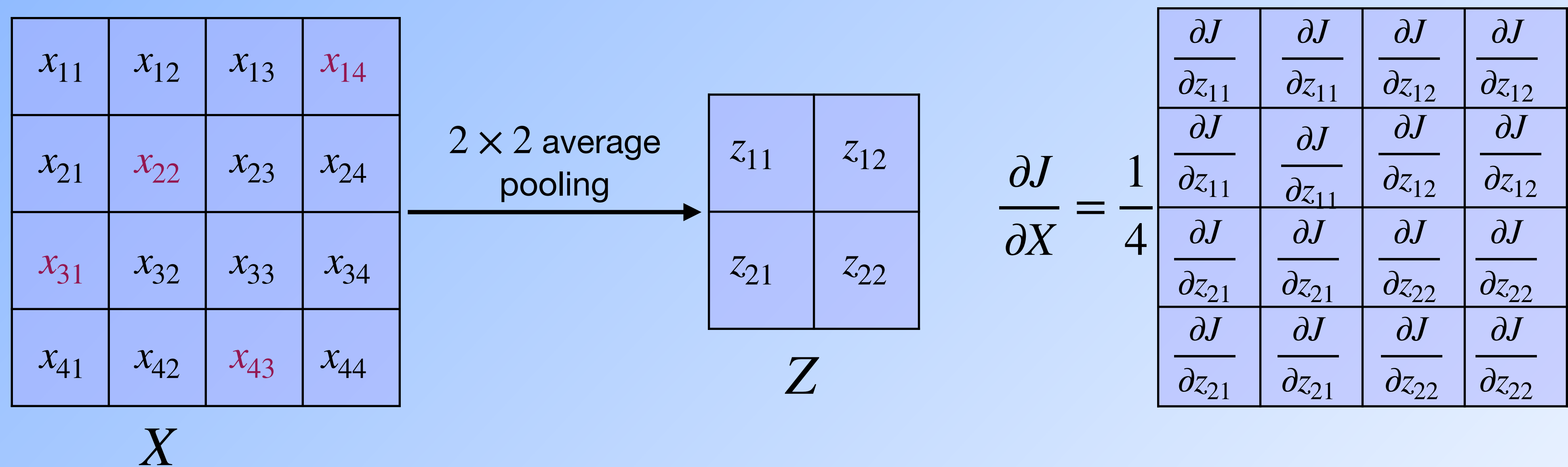
- Spatial pooling layers are often applied to the feature maps produced by the convolutional layers.
- Two types of pooling layers are most prevalent:
 - Max pooling
 - Average pooling
- Pooling layers accomplish the following:
 - Translational invariance
 - Reduction of spatial dimensions of feature maps
 - Faster computations



Backpropagation Through Pooling Layer

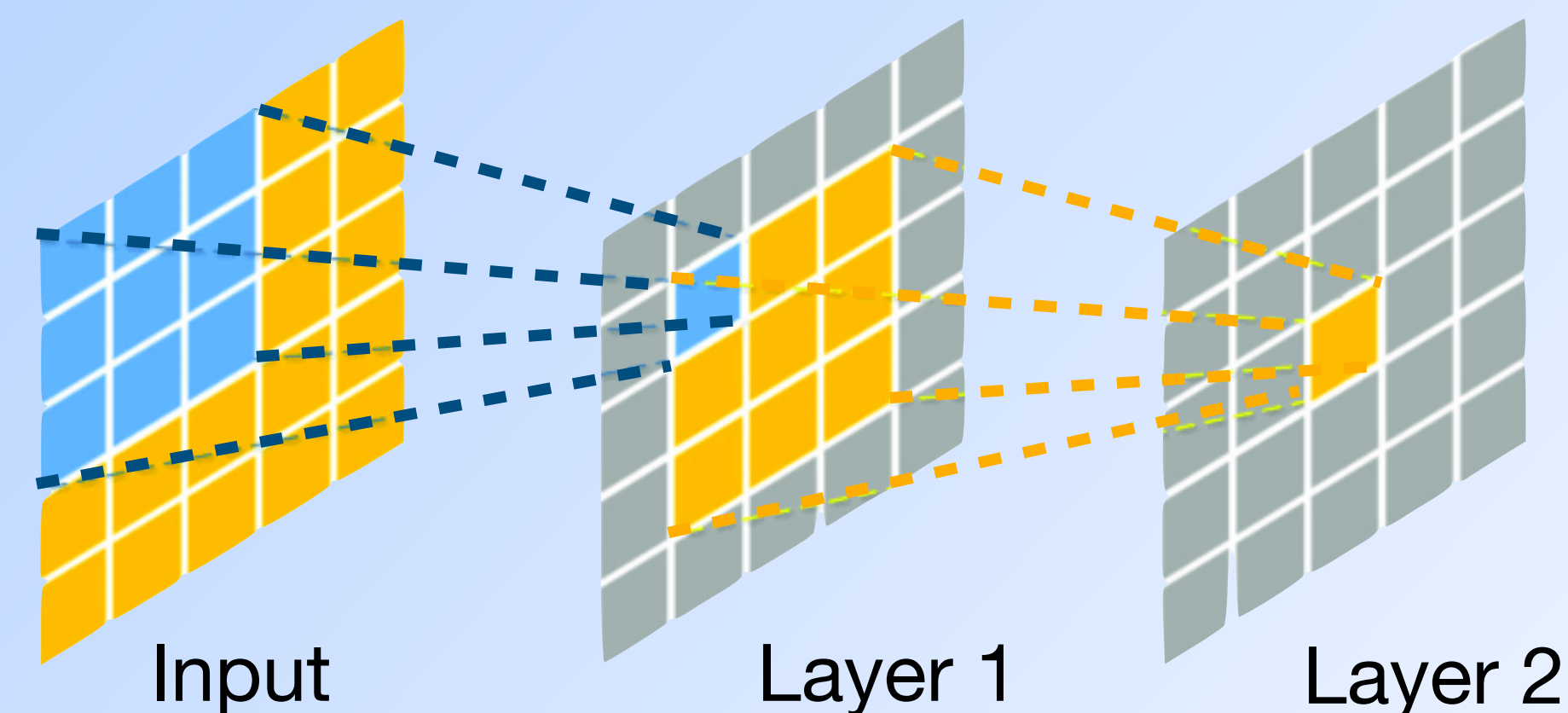


Backpropagation Through Pooling Layer

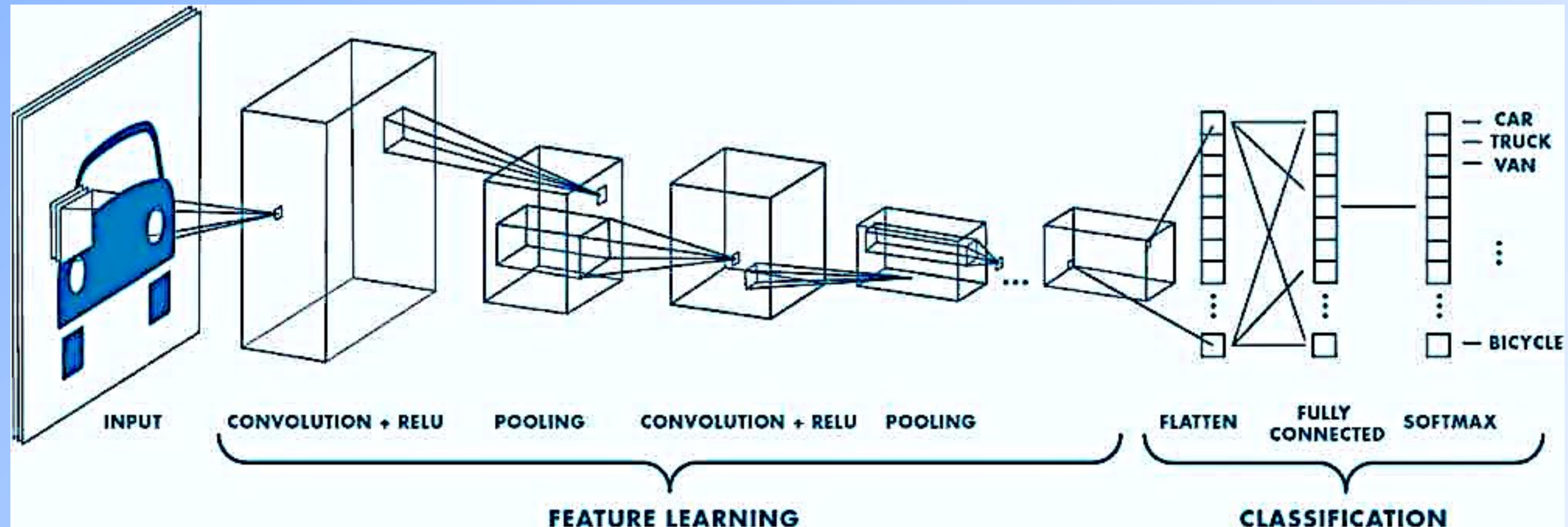


Receptive Field of CNNs

- The receptive field (RF) or field of view (FoV) of a convolutional layer is defined as the area of the input image that is “visible” to this layer.
- In other words, the RF is determined by the pixel area of the input that contributes to a single element of the output of the convolutional layer under consideration.
- The receptive field size for each successive layer of a CNN could be determined by the following recursive formula:
 - For first layer: $R_1 = K_1$, where K_1 is the size of the filter kernel for the first layer.
 - For layer l : $R_l = R_{l-1} + (K_l - 1) \prod_{i=1}^{l-1} S_i$
- Thus, the RF size increases with the depth of the CNN.



Convolutional Neural Network



- Deep feature map produced after several stages of convolutional layers is flattened and passed to a fully connected (dense) layer.
- The predictions are finally obtained by applying an appropriate activation function to the output of the fully connected layer.